INCH-POUND
MIL-M-71229
5 March 1996

MILITARY SPECIFICATION

MOUNT, GUN 120MM

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification supports the acquisition of a hydrospring gun mount for use with the 120MM M256 Gun in the Abrams Series Tanks. This specification includes the minimum essential Engineering and Packaging Requirements and the necessary Quality Assurance Provisions to determine that these requirements have been met.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 1015

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SPECIFICATIONS

MILITARY

MIL-P-14232	- Parts, Equipment and Tools for Army Materiel, Packaging of.
MIL-H-46170	- Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon Base.
MIL-W-63150	- Weapons and Support Material, Standard Quality Assurance Provisions for.
MIL-G-81322	- Grease, Aircraft, General Purpose, Wide Temperature Range, NATO Code Number.

STANDARDS

MILITARY

MIL-STD-129	- Military Marking.
MIL-STD-130	- Identification Marking of U.S. Military Property.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.9)

US Army Armament Research, Development & Engineering Center

B8769067	- Magnetic Particle Inspection of Cannon and Mount Components.
9377499	- Gun Mount Assembly (120mm).
12304935	- Gun Mount Assembly (120mm).

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(Copies of drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Procuring Contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First articles. The contractor shall submit a first article unless it is specifically waived in the contract (see 4.3 and 6.2). No first article requirements shall be waived without review and approval by the Procuring Contracting Officer (see 6.5 and 6.7).

3.1.1 First article sample. Two of the first production gun mounts shall be furnished for first article inspection (see 4.3 and 6.2). These mounts shall be representative of the production processes of the manufacturing facility to be used during quantity production.

3.2 General requirements.

a. The gun mount shall be in compliance with the applicable drawings (9377499 or 12304935), this specification, and all referenced documents. The gun mount shall operate within the design concepts of the system in the initial exercising of the system. There will be no signs of any form, fit or function of the system what will be detrimental to the usage of the system. All materials, finishes, and lubricants shall conform to the specifications noted in the technical data. Certification of the materials, finishes, and lubricants used are acceptable in lieu of tests upon approval of the Procuring Contracting Officer. All components and subassemblies of the gun mount: (1) shall not be altered at assembly; (2) shall be machined, assembled and protected to the extent called out in the technical data; (3) shall show no signs of deterioration or degradation; (4) shall be wiped clean with a lightly oiled lint free rag before being assembled into the gun mount; (5) shall be IAW MIL-W-63150.

b. The gun mount shall be assembled in an area with cleanliness at least equivalent to a machine shop area with no machining activity or construction dust within a distance that could possibly effect the assembly area.

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c. All data and records pertinent to the successful operation of the gun mount must be maintained by the contractor. Any nonconformance to the pertinent technical data shall be resolved according to the order of precedence (see 2.2). Final authority for any approval/ disapproval shall rest with the Procuring Contracting Officer.

3.2.1 Lubrication. When assembling the mount, a grease conforming to MIL-G-81322 shall be applied to each seal assembly, after it has been installed in its groove. Only grease conforming to MIL-G-81322 shall be used to lubricate the internal surface of the mount and the assembled mount.

3.2.2 Hydraulic system. The gun mount shall be filled with hydraulic fluid, bled, and refilled until all air and moisture are eliminated. The initial and subsequent filling of the hydraulic system shall be with hydraulic oil conforming to MIL-H-46170 and filtered through a 5 micron filter. Prior to and after mechanical gymnastication (see 3.3.1), the hydraulic system shall be free of all metallic or abrasive particles that may cause damage to the inner surface of the operating mechanisms. These particles shall not exceed 600 microns in diameter (.024 in) except for nonmetallic lint and fibers, where a maximum of 35 particles between 600 and 3000 microns in greatest dimensions are allowed per one pint sample.

3.2.3 Gun mount static pressure. Prior to proof acceptance testing, each mount shall withstand the sustained hydraulic static pressure of 3500 + 200 psi for not less than five minutes. No leakage permissible. NOTE: Extreme caution must be exercised during pressurization of gun mounts.

3.2.4 Critical components. After proof acceptance testing, each rotor mounting bolt (drawing nomenclature: Screw), king nut (drawing nomenclature: Nut), bearing, adaptor, and thrust nut (drawing nomenclature: Nut) shall not have any indications of cracks or defects.

3.3 Performance. The gun mount, completely assembled and lubricated, shall operate without chatter or binding in all ranges of elevation, depression, and temperatures, without special aids under all operating conditions and as specified herein.

3.3.1 Mechanical gymnastication. Each gun mount shall be subjected to mechanical gymnastication testing at elevations ranging from 0 degrees to 35 degrees with a tolerance of ± 2 degrees as specified in 4.4.2.1 without evidence of erratic movement. A minimum of 20 cycles shall be performed at each elevation tested.

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3.3.1.1 Counter-recoil (CTRR) time. The maximum CTRR time shall be 0.4 seconds at 0 degrees ± 2 degrees elevation, and 2.0 seconds at 35 degrees ± 2 degrees elevation.

3.3.1.2 Hydraulic leakage - mechanical gymnastication. During and immediately following mechanical gymnastication, there shall be no more than 15 drops of oil accumulation at any location in a three minute period.

3.3.2 Proof firing. Each gun mount shall be proof fired (see 6.6.1) or a sample shall be proof fired in accordance with the sampling plan (see 4.4.2.2.2.3) when simulation testing (see 6.6.2) is in effect. Proof firing shall be performed at a Government proving ground and shall comply with the provisions described in 4.4.2.2.1 without evidence of malfunction or failure. For proof firing, the test configuration shall be per the applicable drawing (9377499 or 12304935).

3.3.3 Simulation testing. Simulation testing followed by proof firing according to a sampling plan may be used in lieu of 100% proof firing for proof acceptance testing providing the simulator is qualified as specified in 6.4. Simulation testing for proof acceptance, providing this election is desired and approved, shall be conducted on each gun mount according to the simulation test provisions specified in 4.4.2.2.2 without evidence of malfunction or failure. For simulation testing, the test configuration shall be per the applicable mount assembly drawing.

3.3.4 Recoil cycle. when firing (see 6.6.3) at 100 percent and 113 ± 4 percent acceptance levels (see 6.6.4), the recoil cycle travel (in inches) and total cycle time (in seconds) at 70 degrees F ± 35 degrees F shall be as follows:

3.3.4.1 Recoil travel, in inches.

Elevation in degrees (± 2)	Min - Max (in)
0 to 35	11.25 - 12.25

3.3.4.2 Total recoil cycle time.

Elevation in degrees (± 2)	Max (sec)
0	0.60
35	2.50

3.3.5 Counter-recoil. (CTRR).

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3.3.5.1 CTRR velocity. At 0.50 inch from the complete in-battery position, the CTRR velocity at 70 degrees F ± 35 degrees F shall not exceed the following:

Elevation in degrees (± 2)	Max (ft/sec)
0	3.80
35	3.80

3.3.5.2 Case ejection. During counter-recoil action, the breech operating cam shall operate in conjunction with the breech mechanism to produce ejection of empty cases clear of the cannon and latch breech block in open position at all operational temperatures, except at extreme cold temperatures (-31 degrees F to -50 degrees F). When simulation testing, the velocity of counter-recoil at 3.0 inches ± 0.25 inches from the complete in-battery position shall be as follows (in order to satisfy the case ejection requirement):

Elevation in degrees (± 2)	Min - Max (ft/sec)
0	4.00 - 6.25
35	2.60 - 5.90

3.3.6 Recoil performance zones. Whether 100% proof firing or 100% simulation testing with a sample proof fired, the recoil cylinder oil pressure versus time and piston travel versus time curves for each mount tested (see Table III) shall be within the applicable recoil performance zones shown in Figures 1, 2, 3 and 4 at 70 degrees F ± 35 degrees F.

3.3.7 Hydraulic leakage. During and immediately following each sequence of firing cycles for each gun mount, there shall be no more than 15 drops of oil accumulation at any location in a three minute period. After the mount sets statically for four hours and all joints are wiped clean, there shall be no more than three drops of oil accumulation at any location within a five minute period.

3.3.8 Weapon functioning (visual). Any indications of erratic operations in either recoil or counter-recoil modes shall be cause for rejection.

3.3.9 Climatic conditions. Gun mounts shall be fully operational between the temperature range of -50 degrees F and +125 degrees F without evidence of malfunction or failure. During and after proof firing at extreme climatic conditions, the gun mount shall meet the requirements of 3.3.7 and 3.3.8.

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3.3.9.1 Recoil cycle time. The recoil cycle time under climatic test conditions at 100 percent and 113 ± 4 percent impulse 1eve1s between 0 degrees and 35 degrees elevation shall be as follows:

Temperature (± 5 degrees F)	Max. cycle time (sec)
-50	N/A
-25	25.0
+125	0.6

3.4 Protective finishes.

3.4.1 Inorganic. Inorganic finishes, including metal deposited coatings, and surface treatments shall meet drawing requirements.

3.4.2 Painting. Painting shall be as specified on the drawings.

3.5 Marking. Identification marking, including application of part numbers, and instruction marking shall be as specified on the drawings and in MIL-STD-130 and MIL-STD-129.

3.6 Workmanship. Each mount shall be free of defects that affect appearance, serviceability, or performance such as burrs, fins, sharp edges, and surface flaws or blemishes. Each mount assembly and parts thereof shall meet the requirements of MIL-W-63150.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the

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specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. Unless otherwise specified in the contract (see 6.2), a first article sample of two gun mounts shall be randomly selected from a minimum of four of the first production gun mounts. The first article sample shall be inspected by the contractor and submitted to the Government for approval. Inspection shall be in accordance with all applicable Quality Assurance Provisions to determine conformance with the requirements of this specification. The first article sample shall be representative of the production processes of the manufacturing facility to be used during quantity production.

a. If 100% proof firing will be used for proof acceptance, then all proof firing tests shall be performed.

b. If 100% simulation (with proof firing according to the sampling plan) will be used for proof acceptance, then all simulation tests shall be performed. If the simulator is qualified, no first article proof firing tests are required. If the simulator is not qualified, then all simulation tests and all proof firing tests shall be performed.

4.3.1 First article proof firing. The first article sample shall be proof fired in accordance with 4.4.2.2.1 using Table I in lieu of Table IV. All measurements required by 4.4 of this specification shall be recorded for each mount and each mount shall meet the requirements of 3.3.

4.3.1.1 Climatic conditions. Unless otherwise specified in the contract, the first article sample shall be proof fired to determine compliance with 3.3.9. Proof firing shall be in accordance with Table I at each of the temperatures +125 ± 5 degrees F and -25 ± 5 degrees F. In addition, 5 rounds shall be fired at between 0 degrees and 35 degrees elevation and 100% impulse level at -50 ± 5 degrees F.

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The recoil cycle time shall be measured to determine compliance with 3.3.9.1. Failure to comply with 3.3.9 shall be cause for rejection.

Table I. Proof firing first article sample.

Round number	Elevation (± 2 degrees)	Acceptance level (%) (see 6.6-4)
1	0	50 (approx.)
2	0	75 (approx.)
3	0	100
4	0	113 ± 4
5	10	113 ± 4
6	10	100
7	20	113 ± 4
8	35	100
9	35	113 ± 4
10	35	113 ± 4
11	0	100

4.3.2 First article simulation testing. The first article sample shall be simulation tested in accordance with 4.4.2.2.2 using Table II in lieu of Table V. All measurements required by 4.4 of this specification shall be recorded for each mount and each mount shall meet the requirements of 3.3. When applicable, all simulation tests shall be performed prior to proof firing tests.

Table II. Simulation testing-first article sample.

Round number	Elevation (+ 2 degrees)	Minimum acceptance 1evel (%)
1	0	50
2	0	113
3	0	113
4	35	113
5	35	113
6	35	113
7	35	113
8	35	113
9	35	113
10	0	113
11	0	113
12	0	113
13	0	113

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4.3.3 First article rejection. If any sample fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of the sample to comply with the requirements of this specification.

4.4 Quality conformance inspection. Quality conformance inspection is listed in Table III. Please note that the column labeled "Simulation" represents the minimum inspection frequency if 100% simulation testing (with proof firing according to the sampling plan) is used in lieu of 100% proof firing for proof acceptance.

TABLE III. Quality conformance inspection.

Category/ Characteristic	Requirement Paragraph	Inspection paragraph	Minimum Insp. 100% Proof fire	Frequency * Simulation
Examinations				
General 100%	3.2	4.4.1.1	100%	
Lubrication	3.2.1	4.4.1.1.1	100%	100%
Hydraulic System	3.2.2	4.4.1.1.2	100%	100%
Gun Mount Static Pressure 100%	3.2.3	4.4.1.1.3	100%	
Critical components 100%	3.2.4	4.4.1.1.4	100%	
Protective finish 100%	3.4	4.4.1.1.5	100%	
Marking 100%	3.5, 5.	4.4.1.1.7	100%	
Workmanship 100%	3.6	4.4.1.1.5	100%	
Packaging 100%	5.	4.4.1.1.6	100%	
Tests				
Mechanical gymnastication 100%	3.3.1	4.4.2.1	100%	
Counter-recoil time 100%	3.3.1.1	4.4.2.1	100%	
Hyd. leakage-mech. gymnast. 100%	3.3.1.2	4.4.2.1.1	100%	
Proof firing sample	3.3.2	4.4.2.2.1	100%	
Recoil cycle sample	3.3.4	4.4.2.2.1	100%	

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TABLE III. Quality conformance inspection - Continued.

Category/ Characteristic	Requirement Paragraph	Inspection paragraph	Minimum Insp. 100% Proof fire	Frequency * Simulation
Counter-recoil velocity sample	3.3.5.1	4.4.2.2.1	100%	
Case ejection sample	3.3.5.2	4.4.2.2.1.1	100%	
Recoil performance zones 2/yr	3.3.6	4.4.2.2.1	100%	
Hyd. leakage sample	3.3.7	4.4.2.2.1.2	100%	
Weapon functional (visual) sample	3.3.8	4.4.2.2.1	100%	
Simulation testing	3.3.3	4.4.2.2.2	N/A	100%
Recoil cycle	3.3.4	4.4.2.2.2	N/A	100%
Counter-recoil velocity	3.3.5.1	4.4.2.2.2	N/A	100%
Case ejection	3.3.5.2	4.4.2.2.2.1	N/A	100%
Recoil performance zones	3.3.6	4.4.2.2.2	N/A	100%
Hyd. leakage	3.3.7	4.4.2.2.2.2	N/A	100%
Weapon function (visual)	3.3.8	4.4.2.2.2	N/A	100%

* Proof acceptance may be accomplished by 100% proof firing OR 100% simulation testing with a sample being proof fired. Only the applicable column applies.

4.4.1 Examinations.

4.4.1.1 General examinations. Materials used in the manufacture of mounts shall be examined and tested in accordance with the material specifications referenced on the applicable drawings. Certification provisions of MIL-W-63150 apply. The contractor's inspection records shall be checked to determine conformance to 3.2. Each mount shall be examined to determine compliance with all of the provisions of 3.2. Failure to comply with the requirements of 3.2 shall be cause for rejection.

4.4.1.1.1 Lubrication. Failure to comply with the requirements of 3.2.1 shall be cause for rejection of the mount.

4.4.1.1.2 Hydraulic system. To determine compliance with 3.2.2, a minimum of one pint of oil shall. be drained from an unobstructed access in the lowest position of the hydraulic cylinder of each mount. The sample shall be taken between four and five minutes after mechanical gymnastication. Any contaminant in the hydraulic system must be analyzed to determine whether there is a breakdown in the wiper/seals or if the metallic members are being excessively worn. Failure to comply with the requirements of 3.2.2 shall be cause for rejection.

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4.4.1.1.3 Gun mount static pressure. To determine compliance with 3.2.3, each gun mount shall be pressurized in accordance with 3.2.3 and inspected for damage, permanent deformation, or leakage. Failure to comply with 3.2.3 shall be cause for rejection of the mount.

4.4.1.1.4 Critical components. During and immediately following proof firing or simulation testing, each rotor mounting bolt, king nut, bearing, adaptor, and thrust nut shall be visually inspected for defects. After proof firing or simulation testing, each of these critical components shall be magnetic particle inspected in accordance with B8769067. Failure to comply with the requirements of 3.2.4 shall be cause for replacement of the component. All defects shall be recorded as test observations. All approved critical components shall be stamped by the approving agency.

4.4.1.1.5 Visual examination. Each mount shall be examined for the following defects. one or more defects shall be cause for rejection of the mount.

a. Incorrect assembly or missing parts.

b. Discontinuity or absence of protective finish.

c. Incorrect or missing marking.

d. Color of paint not as specified.

e. Workmanship not as specified.

4.4.1.1.6 Cleaning, drying, preservation, packaging, and packing. Each mount shall be examined to determine compliance with the applicable packaging requirements. Failure to comply with the requirements of section 5 shall be cause for rejection of the mount.

4.4.1.1.7 Marking. marking shall be examined to determine compliance with the requirements of 3.5 and section 5. Failure to comply with these requirements shall be cause for rejection of the mount.

4.4.2 Tests.

4.4.2.1 Mechanical gymnastication. Each mount, including the cannon or simulated equal, shall be mechanically gymnasticated to a recoil displacement of 11.75 $\pm$ 0.5 inches at all required elevations with the gymnasticator set for a 6 $\pm$ 2 second pull back and release time. Each mount shall be subject to a minimum of 40 cycles as follows:

a. 20 cycles at 0 degrees $\pm$ 2 degrees

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b. 20 cycles at +35 degrees ± 2 degrees

Smooth counter-recoil performance is required prior to completion of mechanical gymnastication. The hydraulic system is to be checked in compliance with 4.4.1.1.2 and refilled. Failure to comply with the requirements of 3.3.1 shall be cause for rejection.

4.4.2.1.1 Hydraulic leakage - mechanical gymastication. During and immediately following mechanical gymnastication, the hydraulic leakage shall be checked (# of drops at any location in a three minute period). Failure to comply with the requirements of 3.3.1.2 shall be cause for rejection.

4.4.2.2 Proof acceptance. Mounts shall be proof accepted through 100% proof firing or 100% simulation testing followed by proof firing according to the sampling plan (see 4.4.2.2.2.3). Mounts to be proof fired shall be tested in accordance with 4.4.2.2.1. Failure to comply with the requirements of 3.3.2 shall be cause for rejection. Mounts to be simulation tested shall be tested in accordance with 4.4.2.2.2. Failure to comply with the requirements of 3.3.3 shall be cause for rejection.

4.4.2.2.1 Proof firing. Proof firing shall be as specified in Table IV. Recorded measurements shall be taken on each mount to demonstrate compliance with the requirements of 3.3.4, 3.3.5, and 3.3.7. Recorded measurements shall be taken a minimum of twice a year to demonstrate compliance with the requirements of 3.3.6. After firing, each mount shall be held for observation for four hours and examined to determine compliance with oil leakage requirements specified in 3.3.7. During and after proof firing, each gun mount shall meet the requirements of 3.3.8. Observation shall be made for erratic motion during the entire firing cycle. Failure to comply with the requirements of 3.3.2, and 3.3.4 through 3.3.8 shall be cause for rejection.

Table IV. Proof firing-quality conformance.

Round number	Elevation (± 2 degrees)	Acceptance level (%)
1	0	100
2	0	113 + 4
3	35	113 + 4
4	35	100

4.4.2.2.1.1 Case ejection. During each proof firing cycle, empty cases shall eject properly for each mount proof fired. Failure to comply with the requirements of 3.3.5.2 shall be cause for rejection.

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4.4.2.2.1.2 Hydraulic leakage. Hydraulic leakage rate (# of drops at any location in a three minute period) shall be checked during and immediately following the test sequence specified in Table IV. Failure to comply with the requirements of 3.3.7 shall be cause for rejection.

4.4.2.2.1.3 Proof firing failure. In the event a gun mount fails to satisfy proof firing acceptance criteria, the following procedure will be implemented:

a. The failed mount, accompanied with a report detailing the deficiencies found, shall be returned to the contractor for immediate corrective action to current and subsequent production mounts. No unusual dismantling or altering of the mount will occur at the Government test facility without the documented consent of the contractor.

b. When not simulation testing or when simulation testing is in effect and it is determined that simulation testing is not expected to have revealed the failure, then, after corrective action, further production mounts shall not be released until such time as three additional mounts, selected by the Government, consecutively satisfy acceptance criteria (see 3.3.2) when tested at a Government proving ground.

c. When simulation testing is in effect and if it is determined that simulation testing should have revealed the failure, then, paragraph 4.4.2.2.2.3.b applies.

4.4.2.2.2 Simulation testing. When simulation testing is desired and approved for use as a method of proof acceptance (see 6.4), 100% of the mounts shall be subjected to hydraulic simulation testing with sampling (see 4.4.2.2.2.3) in effect for proof firing. Each mount shall be tested as specified in Table V. Recorded measurements shall be taken on each mount to demonstrate compliance with the requirements of 3.3.4 through 3.3.7. After simulation testing, each mount shall be held for observation for four hours and examined to determine compliance with oil leakage requirements specified in 3.3.7. During and after simulation testing, the gun mount shall meet the requirements of 3.3.8. It is acceptable to perform testing on a test stand at zero elevation with adjustment made as applicable to simulate specified elevations. The simulator shall be varied as required to meet the stated impulse levels. Failure to comply with the requirements of 3.3.3 through 3.3.8 shall be cause for rejection.

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Table V. Simulation testing.

Round number	Elevation (± 2 degrees)	Minimum acceptance level (%)
1	0	Warm-up
2	0	100 (113% UPL)
3	0	100 (113% UPL)
4	35	100 (113% UPL)
5	35	100 (113% UPL)
6	35	100 (113% UPL)
7	35	100 (113% UPL)
8	35	100 (113% UPL)
9	0	100 (113% UPL)
10	0	100 (113% UPL)

NOTES:
(1) The UPL for the M256 cannon (120mm) used with the MlAl gun mount will be: 100% UPL - 88,100 PSI and 113% UPL - 97,000 to 102,000 PSI.

(2) During extremely cold ambient conditions, additional warm-up rounds may be required before record rounds are attempted. Record rounds must be in proper order.

(3) During the conduct of the required record rounds, additional cycles may be included within each operational sequence to achieve compliance with 100% impulse levels and to avoid detectable double bumping or in the event of instrumentation malfunction.

4.4.2.2.2.1 Case ejection. The velocity of counter-recoil at 3.0 inches ± 0.25 inches from the complete in-battery position shall be measured for each mount simulation tested. Failure to comply with the requirements of 3.3.5.2 shall be cause for rejection.

4.4.2.2.2.2 Hydraulic leakage. Hydraulic leakage rate (# of drops at any location in a three minute period) shall be checked during (if there is a break of more than three minutes between firing cycles), immediately following, and four hours after the test sequence specified in Table V. Failure to comply with the requirements of 3.3.7 shall be cause for rejection.

4.4.2.2.2.3 Sampling. Sampling for proof firing shall be in accordance with the sampling requirements specified in the acquisition documents (see 6.2), the Gun Mount Assembly Drawings (9377499 and 12304935), and this specification. The sampling plan shall include the following sampling provisions:

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a. The initial or "i" quantity of mounts (as specified in the acquisition documents or in the drawings) shall be simulation tested in accordance with 4.4.2.2.2 and proof fired in accordance with 4.4.2.2.1. When the "i" quantity has been satisfied then 100% of the mounts shall be simulation tested and sampling allowed for proof firing according to a frequency rate or "f".

b. If any mount failures occur during proof firing and it **is** determined that simulation testing should have revealed the failure, then the following procedures apply: corrective action shall be taken (including any necessary modification of simulation in order to reveal the failure in the future), all mounts produced since the previous accepted mount via proof firing shall be proof fired, and an additional "i" quantity shall be both simulation tested and proof fired.

4.4.2.3 Records. The contractor shall prepare a Quality Inspection Test, Demonstration and Evaluation Report giving the detailed test results for 4.4.2.1 and 4.4.2.2 (see 6.2, 6.3).

4.5 Inspection-equipment. All inspection-equipment shall be in accordance with MIL-W-63150 and MIL-I-45607, as applicable.

5. PACKAGING

5.1 Pilot pack. A pilot pack consisting of a complete and packed shipping container, packaged in accordance with the applicable packaging requirements (see 5.2) to the level of protection specified in the contract (see 6.2), shall be forwarded in compliance with the contract.

5.2 Cleaning, drying, preserving, packaging, packing, and markings.

5.2.1 Mounts. The mounts shall be cleaned, dried, preserved, packaged, and packed. Marking shall be as specified in MIL-STD-129, as applicable (see 6.11).

5.2.2 Repair parts. Repair parts shall be cleaned, dried, preserved, unit packed, packed, and marked in accordance with the applicable packaging requirements (see 5.2.1) for the level of protection specified in the contract (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. This type of hydrospring gun mount is designed for use with the 120MM: M256 Tank Gun in the Abrams Series Tanks.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for and the number of mounts, components hereof, or both to be submitted for first article by the contractor (see 3.1, 4.3, 6.5).

d. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

e. All inspection and test data that is required.

f. At whose expense a retest may be performed.

g. Any additional or extended examinations and tests beyond the scope of this specification.

h. Serial numbers for the gun mounts.

i. Inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5).

j. Availability of inspection equipment from the Government.

k. Extent of contractor's responsibility for Government furnished and for contractor-required final inspection equipment.

1. Applicable acceptance test procedures.

m. When a proof firing sampling plan is authorized, the proof firing sampling plan parameters: initial consecutive quantity (i), and sampling frequency (f) (see 4.4.2.2.2.3).

n. Under what conditions simulated firing may be performed, the test equipment available, and the procedures to be followed.

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o. The levels of preservation, packaging, and packing required for mount assemblies (see 5.2.1) and repair parts (see 5.2.2).

p. Marking and bar code marking requirements (see 5.2).

q. When warranted, the application of MIL-Q-9858 or MIL-1-45208, as appropriate.

r. Unless otherwise specified, the application of MIL-1-45607 and MIL-STD-45662.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 227.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
4.4.2.3	DI-R-1724	Quality Inspection Test Demonstration and Evaluation Rpt.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Simulation testing. General information regarding mechanical gymnasticators or hydraulic simulators is available from the Commanding Officer, ATTN: SMCRI-APA, Rock Island Arsenal, Rock Island, IL 61299-5000. Simulators may be qualified through utilization with two mounts successfully meeting the requirements of first article inspection and through utilization with the "i" quantity of mounts successfully meeting the requirements of quality conformance inspection. Simulation data must correlate with fully instrumented proof firing data as approved by the Engineering Design Agency. Simulation/sampling shall not be used for proof acceptance prior to written approval from the Project Manager, Abrams Tank System.

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6.5 First article. When first article inspection is required, the Procuring Contracting Officer should provide specific guidance to offerers whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first lot of production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3. The Procuring Contracting Officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Definitions.

6.6.1 Proof firing. For purposes of clarity, proof firing is the term used when reference is made to actual live firing only.

6.6.2 Simulation testing. For purposes of clarity, simulation testing and simulated firing cycles are terms used when reference is made to simulation only.

6.6.3 Firing. For purposes of clarity, firing and firing cycles are terms used when reference is made to both proof firing and simulation testing.

6.6.4 Acceptance level. The 113% acceptance level for testing of Abrams Tank gun mounts shall be defined as the anticipated maximum impulse generated by a special 113% upper pressure limit (UPL) (97,000 to 102,000 psi) proof charge consisting of a 120mm super slug projectile (28.8 lbs) assembled to the combustible case with an adjusted charge weight of approximately 17.90 lbs of M30 MP propellant conditioned for 24 hours at temperatures up to 145 degrees F. The anticipated nominal muzzle velocity for this proof charge is 4,500 ft/sec $\pm$ 20 ft sec with a minimum delivered total impulse of 7374 lb-sec. For purposes of simulation testing, the acceptance level applied to the gun mount shall be computed from the measured area under the recoil cylinder oil pressure vs time data curves plus the energy dissipated within the counter-recoil spring and system frictional loads.

NOTE: Similarly, 100% UPL is approximately 88,100 psi.

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6.7 Waivers and deviations. The Procuring Contracting Officer shall coordinate all requests for waiver or deviation to this specification with Benét Laboratories and the Product Assurance Directorate at Picatinny Arsenal.

6.8 Subject-term (key word)-listing.

Hydrospring
M1Al tank

6.9 Drawings. Drawings listed in section 2 of this specification under the heading "U.S. Army Armament Research, Development and Engineering Center" (ARDEC) may also include drawings prepared by, and identified as Watervliet Arsenal, Ordnance Corps, weapons Command, or U.S. Army Armament Research and Development Command, U.S. Army Armament, Munitions and Chemical Command, etc. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.10 Changes from previous issue. None. This is a new specification.

6.11 Packaging and marking. Historically, MIL-STD-1190 has provided the requirements for the cleaning, drying, preservation, and packaging and MIL-STD-1189 has provided the requirements for bar code symbology for these gun mounts.

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Preparing Activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-M-71229	**2. DOCUMENT DATE *(YYMMDD)*** 960305

3. DOCUMENT TITLE

MOUNT, GUN 120MM

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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